Local competition and pizza prices

Group 9 – Uber Eats listings around Houston

1. Question and expected relationship

Do restaurants with more nearby pizza competitors list lower pizza prices, after accounting for location, identified chains and menu characteristics?

Customers can switch to nearby sellers, which may limit the prices a restaurant can charge. We therefore initially expect a negative relationship. However, areas with high demand may attract more restaurants and also support higher prices. The sign observed in a cross-section is not necessarily the effect of a new competitor entering the market.

Our main model gives a small positive association, about **+0.59% per additional nearby site**. We explain how the data and controls lead to this result and why it cannot be interpreted as a causal effect.

2. What we measure

We use the public *Uber Eats USA Restaurants and Menus* dataset [1]. The sample contains **597 listings representing 552 sites**, within 50 km of central Houston, at coordinates (29.7604, −95.3698). One listing is one platform record, not necessarily one separate physical restaurant.

Price. For each listing, we average the prices of retained whole-pizza offers, giving each offer equal weight. Slices, bundles, kits, sides, nonpositive prices and explicit rectangular formats are excluded. Repeated identical offers are removed. Explicit diameters must be 8–18 inches; offers with unknown diameters remain. This is a listed menu price, not a sales-weighted or transaction price.

Competition. We count distinct other pizza-selling sites within **2 km**, using great-circle distances and grouping coordinates rounded to four decimals. We exclude the whole own site. Suppliers with missing prices or ratings remain counted. A 55 km supplier area provides coverage around the 50 km study boundary.

Sample feature	Value
Pizza listings before selection on an eligible price	733
Listings with an eligible price used in all models	597
Mean listed pizza price	$14.85
Mean number of other pizza sites within 2 km	5.17
Mean share of offers with a known diameter per listing	21.2%

The 136 listings without an eligible price are excluded from the regressions, but remain potential competitors. Their mean competition count is 6.10, compared with 5.17 in the final sample. Price availability may therefore select the sample.

3. Four models with a clear sequence

We estimate the same 597 listings in every model. The outcome is the natural logarithm of mean pizza price. For M1–M4:

$$\log(P_i) = \alpha + \beta\,\text{Competition}_i + \text{controls}_i + u_i.$$

The coefficient β compares listings with different competitor counts while holding the included characteristics fixed. Using the log of price makes percentage comparisons convenient; it does not make the estimate causal.

Model	What is added	Why
M1	Competition only	Describe the unadjusted association
M2	Location controls	Account for broad geographic price differences
M3	Identified-chain indicator	Account for observable differences between brand groups
M4	Menu characteristics	Account partly for different pizza sizes and menu composition

Location controls are east and north coordinates in 10 km units, their squares and their product. They allow a smooth geographic pattern; they are not neighborhood fixed effects.

Identified chain is a name-based indicator covering recognized national and local brands. The other group is "other or unclassified", not verified independent restaurants.

Menu controls are mean known diameter, a dummy for entirely missing diameters, the known-diameter share, and the shares labeled small and large. Mean diameter is centered at 12 inches and set to zero after centering when unknown; the missingness dummy distinguishes those cases. Small and large labels are not converted into invented diameters.

4. How much does the fit improve

Model	$\widehat{\beta}$	HC3 SE	Adjusted R^2	Test RMSE (USD)
M1	0.00665	0.00246	0.009	4.62
M2	0.00548	0.00265	0.020	4.54
M3	0.00562	0.00264	0.022	4.60
M4	0.00584	0.00234	0.226	3.91

Table 1. HC3 standard errors allow unequal error variances. Adjusted R^2 accounts for the number of regressors. Test RMSE is a dollar prediction-error measure; smaller is better.

The largest change occurs when menu controls are added: adjusted R^2 rises from 0.022 in M3 to **0.226 in M4**. The competition slope changes relatively little. Explaining more price variation is not the same as changing the answer about competition.

For an internal prediction check, we fit on 477 listings and evaluate on 120 held-out listings, using one fixed random split. Dollar RMSE falls from \$4.62 in M1 to \$3.91 in M4. Dollar predictions use a correction calculated from training residuals when converting fitted log prices back to dollars. Nearby listings or related brands may appear in both sets, so this is a limited internal check.

5. Main result and four tests

M4 estimates $\widehat{\beta} = 0.005841$ with HC3 standard error 0.002341. For one additional observed pizza site within 2 km, the fitted percentage comparison is

$$100(\exp(\widehat{\beta}) - 1) = +\mathbf{0.586}\%.$$

Its 95% interval is **+0.12% to +1.05%**. Five additional sites correspond to about +2.96%. These are comparisons of exponentiated fitted log prices, often called fitted geometric prices. They are not automatically percentage changes in the arithmetic mean price.

We use four explicitly defined tests. T1 is a two-sided t test; T2–T4 are HC3 Wald F tests. We compare the reported nominal p values with 5%; these are approximate tests under their assumptions.

Test	Model	Null hypothesis	Statistic	p
T1	M4	Competition coefficient = 0	$t = 2.495$	0.0129
T2	M4	All 11 control coefficients = 0	$F = 22.444$	< 0.0001
T3	M4	All 12 slopes = 0	$F = 22.068$	< 0.0001
T4	M5	Chain slope difference = 0	$F = 0.572$	0.4499

Table 2. T2 has 11 restrictions (5 location, 1 chain and 5 menu terms); T3 has 12, including competition. T4 has one restriction. These are different questions, not four independent confirmations of a causal effect.

T1: competition. Reject a zero competition slope in M4. The estimated association is positive, contrary to our initial negative expectation.

T2: added controls. Reject that all 11 control coefficients are zero. Together, they add explanatory information relative to M1; this does not prove that each is individually necessary.

T3: overall model. Reject that all 12 M4 slopes are zero. This supports some joint explanatory content, not a correct model in every respect.

6. One extension: does the slope differ for chains

Our only extension, **M5**, adds competition × identified chain to M4. Both main effects remain:

$$\log(P_i) = \alpha + \beta C_i + \theta D_i + \delta C_i D_i + \text{other M4 controls}_i + u_i.$$

When $D_i = 0$, the competition slope is β; when $D_i = 1$, it is $\beta + \delta$. The interaction δ tests the difference between the two slopes.

M5 group	Per-site comparison	95% interval
Other or unclassified	+0.70%	[+0.14%, +1.27%]
Identified chain	+0.30%	[-0.58%, +1.19%]

T4: difference between groups. With $p = 0.450$, we do not establish different slopes. A significant slope in one group and an insignificant one in another would not by itself establish a difference. The interval for $\beta + \delta$ uses their covariance, rather than adding standard errors. M5 does not improve adjusted R^2 or dollar RMSE relative to M4, so we keep M4 for the main comparison.

7. What still limits the model

Two checks on M4

Check	What it asks	p
Breusch–Pagan	Is the error variance constant?	3.53×10^{-6}
RESET	Do added squared and cubed fitted values improve the equation?	0.0058

Breusch–Pagan rejects constant variance under its independent-observation reference. This motivates HC3 rather than homoskedastic standard errors. **RESET** rejects the tested M4 specification using HC3 inference. Some relevant features of price may still be missing or poorly represented. RESET does not tell us precisely which variables to add.

These checks are separate from T1–T4: they assess error variance and the specification, rather than whether selected coefficients are zero. HC3 does not correct spatial dependence; nearby restaurants may share unobserved shocks. This limits the uncertainty assessment as well as the interpretation of independence-based checks.

Pizza sizes and missing information

Only 21.2% of retained offers have a known diameter *on average across listings*. This is not necessarily the pooled share of all offers. We extract diameters when specified in menu text; we do not infer them from prices. Unknown-size offers remain, and the controls cannot make every menu fully comparable.

As one direct check, requiring at least half the offers to have a known diameter leaves **111 listings**. The estimated association is about +0.40%, with an interval containing zero ($p = 0.119$). The result is less precise in this stricter sample. This may reflect sample size and selection as well as product comparability; it does not establish a different true effect.

Demand and entry are not separated

More attractive areas may have higher rents, stronger demand, more expensive products and more competitors. Expected profits can also attract entry. Our data do not isolate competition from these factors. Controlling menu characteristics changes the comparison and may also remove some product differences linked to competition.

Counts cover observed Uber Eats suppliers, not all restaurants. The menu data are a retained regional excerpt whose completeness against the national file is not independently established. Prices are listed offers, not actual purchases. More controls, a higher R^2 and robust standard errors do not remove these limitations.

8. Conclusion

We do not find the expected negative conditional relationship. M4 gives a small positive association, about **+0.59% per additional observed site within 2 km**. Menu controls substantially improve fit, while the chain interaction adds no clear evidence of different slopes.

This is a useful descriptive result, with remaining measurement and specification problems. It does **not** show that competition causes higher prices. A causal answer would require a credible change in competition that is separate from changes in local demand.

Source and reproduction

[1] Ahmed Shahriar Sakib, *Uber Eats USA Restaurants and Menus*, Kaggle: `https://www.kaggle.com/datasets/ahmedshahriarsakib/uber-eats-usa-restaurants-menus`. Source files dated 22 November 2023, not verified individual collection dates. The supplied restaurant file and retained regional raw-menu excerpt are included in the project. The reading guide links each table to its numerical output. Run `reproduire.py` to rebuild the analysis and this report.